UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS

Doctoral Program in Deep Tech, Artificial Intelligence & Emerging Technologies

Research Methods and Scientific Integrity in AI and Advanced Technologies

RESEARCH PROTOCOL [INITIAL DRAFT]

DELIVERABLE No. 3

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2026 - II

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DELIVERABLE No. 3

Research topic: A Verifiable Generative Artificial Intelligence Model for Contractual Clause Recommendation in Regulated Fiduciary Institutions

Protocol note. This deliverable presents the full ten-component protocol structure reviewed in class and carries forward the paradigm statement (Deliverable 1) and the method-fit decision (Deliverable 2), including the revisions requested in the instructor's feedback. Sections 02 (Abstract) and 04 (Literature Review) are preliminary and will be completed after the PRISMA-based systematic review. Institution-specific figures will be reported only in aggregated, de-identified form.

01. Title

A Verifiable, Human-in-the-Loop Generative AI Model for Contractual Clause Recommendation in Regulated Fiduciary Institutions

The title refines the research topic stated in Deliverable 1 by making explicit two commitments already adopted in Deliverables 1 and 2: the artefact is advisory and operates under human authority (human-in-the-loop), and its central quality is verifiability, understood as the traceability of every recommendation to a contrastable normative or contractual source.

02. Abstract

Student-authored abstract scaffold. In the final version, this section will synthesise the problem, the specific research gap, the objective of the verifiable clause-recommendation artefact, the Design Science Research approach grounded in critical realism, the bounded evaluation context and primary outcomes, and the expected scientific contribution. It will be completed after the literature review and the refinement of the evaluation design so that it reflects the final study.

- **Background / problem:** generative AI can draft fluent contractual language, but in regulated fiduciary institutions an unverifiable or hallucinated clause is a legal and supervisory risk, and retrieval-augmented generation (RAG) alone has not eliminated hallucination in legal tools.

- **Research gap:** limited evidence on how to design and evaluate clause-recommendation systems whose every output is attributable to identified, temporally valid Peruvian normative or contractual sources, in Spanish, under SBS supervision.
- **Objective and method:** design and evaluate one bounded artefact under the DSR cycle of Peffers et al. (2007), grounded in critical realism.
- **Evaluation focus:** attribution quality, hallucination rate, and legal correctness / recommendation quality, compared against no-retrieval, standard RAG, and current-practice baselines.
- **Expected contribution:** an evaluated artefact plus nascent, transferable design principles for verifiable generation in regulated domains (Gregor & Hevner, 2013).

03. Introduction & Problem Statement

Fiduciary institutions draft and review contracts whose clauses must conform to a layered regulatory framework. In Peru, trusts are governed by Law No. 26702, the General Law of the Financial and Insurance Systems, and by the Regulation on Trusts and Fiduciary Service Companies, approved by SBS Resolution No. 1010-99, as amended, under the supervision of the Superintendencia de Banca, Seguros y AFP (SBS). Clause drafting in this setting is knowledge-intensive: lawyers rarely draft from scratch; instead, they locate and adapt precedent clauses and check them against the applicable norms (Wang et al., 2025). As of 31 July 2026, the three specialised fiduciary service companies supervised by the SBS administered trusts with combined assets of approximately S/ 78.1 billion (about US$ 23.0 billion at the SBS accounting exchange rate of S/ 3.396), a figure that excludes trusts administered by banks and other financial-system firms authorised to act as trustees (SBS, 2026).

Generative large language models (LLMs) promise to accelerate this work, yet they produce fluent text that may be unsupported by any source. Legal hallucination is now well documented (Dahl et al., 2024). Retrieval-augmented generation (Lewis et al., 2020) is commonly presented as the remedy, but a preregistered evaluation of leading commercial legal research tools built on RAG found that they still hallucinated between 17% and 33% of the time (Magesh et al., 2025). In clause drafting specifically, a recent evaluation of lightweight LLMs across 22 clause categories found that most generated clauses looked professionally drafted while still requiring legal review or substantive correction (Narayanan & Singh, 2026).

In a fiduciary institution, an unsupported clause is not merely a quality defect; it is a source of legal, supervisory, and reputational exposure.

The national policy context reinforces the problem. Ley N.º 31814 and its Regulation, approved by Decreto Supremo N.º 115-2025-PCM (in force since 22 January 2026), establish guiding principles for the development and use of AI systems, including human oversight and personal-data protection, together with a risk-based classification of AI uses. Any clause-recommendation system deployed in a supervised entity must therefore be able to show *why* each output should be trusted and must remain under human authority.

The scientific problem is therefore not whether generative AI can draft clauses, but how a generative artefact can be designed so that every recommendation is verifiable, that is, traceable to a contrastable normative or contractual foundation, without loss of legal quality. Verifiability, in this thesis, is the technical translation of the principle of refutability: a recommendation that cannot be traced to an identifiable source cannot be contested, and therefore cannot be trusted in a regulated domain. Current evidence does not establish how such an artefact should be designed and evaluated for Spanish-language contracts governed by Peruvian fiduciary regulation. In the host institution, the documented as-is process is template-based: once due diligence and management approval are complete, a legal analyst assigned according to the trust type drafts the contract from typology-specific templates; the draft then goes through review cycles with the client, in which most observations concern the negotiation of specific clauses. Notably, the documented process contains no explicit step in which each clause is verified against the applicable normative sources; regulatory conformity rests on the templates and on the individual lawyer's judgement.

To remain feasible within thirty-six months, the study is bounded to one functional prototype, one SBS-supervised fiduciary institution, one clearly delimited clause-recommendation task, and two to three primary outcomes. The study seeks rigorous evaluative evidence and transferable design knowledge rather than claims of universal legal correctness.

04. Literature Review

Preliminary evidence map. Five recent studies (2025–2026) are incorporated at this stage because they directly inform the technological and evaluation space of the thesis. Two are preprints and are flagged as such; their findings will be re-examined against peer-reviewed versions during the systematic review.

Table 1

Recent Evidence on Legal Hallucination, Attribution, and Contract-Clause Generation Relevant to the Proposed Thesis

Study	Design / sample	Main finding	Gap / relevance to this thesis
Magesh et al. (2025)	Preregistered evaluation of RAG-based commercial legal research tools (Lexis+ AI, Westlaw AI-Assisted Research, Ask Practical Law AI) and GPT-4 on real legal research queries, scored by human raters.	RAG reduced hallucinations relative to a general chatbot, but each tool still hallucinated in 17–33% of responses; the study also proposes a typology separating hallucinations from accurate responses.	Shows that RAG alone does not guarantee verifiability and justifies standard RAG as the primary baseline. It addresses US case-law research, not contract drafting, Spanish-language sources, or Peruvian fiduciary regulation.
Wang et al. (2025) — ACORD	First expert-annotated benchmark for contract-clause retrieval: 114 queries and over 126,000 query–clause pairs graded 1–5 by experts.	Bi-encoder retrieval with LLM re-rankers is promising, but substantial improvement is still needed for real drafting work.	Confirms precedent retrieval as the foundation of drafting and offers a model for expert-graded relevance in the gold standard. It covers retrieval only (no generation or attribution) and English-language commercial contracts.
Wallat et al. (2025)	Conceptual and experimental study of citations in RAG; separates citation correctness from citation faithfulness and probes post-rationalisation.	Up to 57% of citations were not faithful: the model cited a supporting document it had not actually relied on.	Directly sharpens the thesis' central tension: citation precision and recall measure correctness, not faithfulness. The evaluation must add a faithfulness probe or state this limit explicitly.
Ju & Lee (2026) — preprint	Regulatory compliance question answering over multi-tier authority structures; proposes citation-closure retrieval with per-rule attribution and a dedicated benchmark.	Native RAG showed an eager-citing bias when hard negative passages were present, with a sharp fall in citation precision; even frontier models were susceptible.	The layered structure law → regulation → circular mirrors the Peruvian SBS framework; supports per-rule attribution as a design candidate. Not contract drafting; jurisdiction differs from Peru.
Narayanan & Singh (2026) — preprint	Risk-sensitive evaluation of four lightweight LLMs across 22 clause categories and 34 legal failure modes; LLM-council scoring with sampled human review.	About 59% of generated clauses fell in intermediate risk levels requiring legal review or correction; proposes a Max-Severity rule so a single decisive defect is not averaged away.	Supports evaluating legal correctness separately from attribution with a severity-based rubric. The authors note it is not anchored to any jurisdiction or to financial-sector regulation, and used no retrieval or attribution layer.

Note. Author's own elaboration based on Magesh et al. (2025), Wang et al. (2025), Wallat et al. (2025), Ju and Lee (2026), and Narayanan and Singh (2026). The table synthesises design, sample, main findings, and gaps relevant to the proposed thesis.

- The recent evidence strengthens the rationale for verifiable generation, but it also **narrows the novelty claim**: the thesis should not claim novelty merely because it uses RAG, citations, or LLMs for contracts, all of which exist in the literature.
- The more defensible contribution is the **integrated design problem**: a Spanish-language, human-in-the-loop clause-recommendation artefact whose outputs are attributable at proposition level to **temporally valid** Peruvian normative and contractual sources, evaluated jointly on attribution, hallucination, and legal correctness in an SBS-supervised setting.
- Two findings shape the design directly: attribution correctness is not faithfulness (Wallat et al., 2025), and an attributable clause may still be legally deficient (Narayanan & Singh, 2026). Both confirm that verifiability is a necessary but not sufficient condition for a trustworthy recommendation.
- ▪ This section will be extended through a PRISMA 2020 systematic review (Page et al., 2021) to substantiate the gap claim, covering Scopus, Web of Science, IEEE Xplore, the ACL Anthology, and SSRN, complemented by a review of SBS regulation and Peruvian State AI initiatives. The search string combines three concept blocks: (1) generative models: ("large language model*" OR LLM OR "generative AI" OR "retrieval-augmented generation" OR RAG); (2) verifiability: (attribut* OR citation* OR verifiab* OR hallucinat* OR grounding OR faithfulness OR traceab*); and (3) legal domain: (contract* OR clause* OR "legal drafting" OR regulat* OR fiduciar*). Records published between January 2020, when retrieval-augmented generation was introduced (Lewis et al., 2020), and the date of execution, in English or Spanish, will be eligible. Inclusion and exclusion criteria will be applied in two screening stages (title/abstract and full text), and record counts at each stage will be reported in a PRISMA flow diagram.

05. Research Questions, Hypotheses & Objectives

Main research question: How can a verifiable, human-in-the-loop generative AI artefact be designed and rigorously evaluated to recommend contractual clauses whose every output is traceable to a contrastable normative or contractual source, in one SBS-supervised fiduciary institution and one clearly bounded clause-recommendation task, and does it outperform a standard retrieval-augmented generation (RAG) baseline on verifiability and hallucination control without loss of legal quality?

Table 2
Research Questions of the Proposed Study

Code	Research question
RQ1: Design	What design principles govern a verifiable clause-recommendation artefact whose outputs are attributable, at proposition level, to identified and temporally valid normative or contractual sources in an SBS-supervised fiduciary setting?
RQ2: Evaluation	Does the artefact achieve higher attribution quality and lower hallucination rates than (a) a no-retrieval baseline, (b) standard RAG, and (c) current institutional practice, while remaining non-inferior in legal correctness and recommendation quality?
RQ3: Risk & ethics	What confidentiality, over-reliance (automation bias), outdated-source, and accountability risks emerge during use, and what safeguards are required to manage them responsibly?

Note. Author's own elaboration. The research questions are organised into three dimensions: artefact design, evaluation, and risk and ethics.

Evaluation hypotheses. Because DSR is governed by design-oriented questions, hypotheses are limited to the quantitative evaluation of the artefact. Conditions B0–B3 are defined in Section 06. Comparisons with current institutional practice (B3) are treated as operational and exploratory.

Table 3
Testable Hypotheses for the Evaluation Phase

Code	Testable hypothesis
H1	On the same evaluation items, the proposed artefact (B2) will achieve higher citation precision and citation recall (Gao et al., 2023) than standard RAG (B1).
H2	On the same evaluation items, the proposed artefact (B2) will produce a lower proportion of hallucinated propositions (unsupported or misattributed) than standard RAG (B1) and the no-retrieval baseline (B0).
H3 (non-inferiority)	The proportion of drafting queries for which at least one suggested clause is judged legally acceptable by the expert annotators will not be lower for the proposed artefact (B2) than for standard RAG (B1) by more than a prespecified non-inferiority margin Δ, expressed in percentage points. Δ will be fixed by structured consensus of the legal experts before data collection and recorded in the preregistration.

Note. Author's own elaboration. H3 is a non-inferiority hypothesis: failing to find a difference does not establish equivalence, so the margin must be fixed in advance.

General objective: By the end of the thirty-six-month doctoral period, design, develop, and rigorously evaluate a verifiable, human-in-the-loop generative AI artefact that recommends contractual clauses traceable to contrastable normative or contractual sources within one SBS-supervised fiduciary institution, while preserving confidentiality, legal quality, and human authority.

Specific objectives:

- Substantiate the research gap through a PRISMA-based systematic review and identify the functional, regulatory, and verifiability requirements of the artefact with the institution's legal area.
- Design and implement the artefact, integrating a de-identified precedent repository, a versioned normative corpus, retrieval, generation with a self-hosted model, a proposition-level verifiability layer, and a human review interface.
- Build an expert-annotated gold standard with reported inter-annotator agreement and evaluate the artefact against baselines B0–B3 on attribution, hallucination, and legal correctness.
- Assess confidentiality, over-reliance, outdated-source, and accountability risks, and refine the artefact in response to evidence.
- Derive and validate, with an expert panel from other supervised entities, nascent design principles for verifiable generation in regulated domains.

06. Methodology

Philosophical position and research approach. The study is grounded in **critical realism** (Bhaskar, 2008) and uses Design Science Research as the governing approach (Hevner et al., 2004). Critical realism distinguishes the real (generative mechanisms such as the regulatory framework, the legal logic of clauses, and model behaviour), the actual (events such as a generated clause or a hallucination), and the empirical (what can be observed and measured). Pragmatism, the more common underlabourer of DSR, is not adopted because verifiability presupposes sources whose content exists independently of the model, the user, or the researcher; the pragmatic utility criterion is nonetheless retained at the methodological level as evidence about the underlying mechanisms, not as the final arbiter of truth.

DSR process. The study follows the six activities of Peffers et al. (2007): (1) problem identification and motivation; (2) definition of solution objectives; (3) design and development; (4) demonstration in the target context; (5) evaluation; and (6) communication. Evaluation episodes are planned with the FEDS framework (Venable et al., 2016), moving from formative, artificial evaluation (benchmarking on the gold standard) to summative, naturalistic evaluation (use by legal officers on real, de-identified drafting tasks).

Researcher positionality. The researcher holds a technical leadership role in the host institution. This insider position facilitates access but introduces risks of designer–evaluator

bias and conflict of interest. Action Design Research (Sein et al., 2011) was considered because of its organisational intervention logic; DSR is retained because the primary contribution is the artefact and its design principles, while the following safeguards are adopted: annotators are blind to the system condition, hypotheses and analysis are preregistered, legal judgements are made by external qualified annotators, and the researcher does not annotate.

Artefact. The artefact is a Spanish-language clause-recommendation assistant embedded in the legal analyst's drafting task. The analyst types the clause need in natural language; the interface autocompletes the request from the knowledge base and returns a small set of candidate clauses, typically two or three, each accompanied by citations to the precedent clauses and normative provisions that support it. The artefact comprises seven components: (i) a de-identification pipeline applied before any processing; (ii) a knowledge base of de-identified precedent clauses and applicable norms (Law No. 26702, SBS Resolution No. 1010-99, and relevant circulars), periodically synchronised from its source systems and versioned with the validity dates of each provision; (iii) query autocompletion over this knowledge base; (iv) hybrid retrieval of precedents and norms; (v) generation with a self-hosted model; (vi) a verifiability layer that decomposes each candidate clause into propositions, checks that each proposition is supported by the cited span and that the cited provision is in force, and withholds any candidate lacking sufficient support; and (vii) a human review interface in which the analyst accepts, edits, or rejects each suggestion. Suggestions are advisory: no clause is adopted without the analyst's decision. For evaluation purposes, the knowledge base is frozen at a dated snapshot so that results remain reproducible despite ongoing updates.

Bounded task and data. The clause-recommendation task is restricted to one clause family within one trust type (e.g., guarantee, cash-flow, or administration trusts), to be selected with the institution's legal area during the requirements phase according to three criteria: frequency of use, degree of regulatory dependency, and legal risk. Data comprise the de-identified precedent corpus drawn from the institution's typology-specific templates and executed contracts, whose size will be reported only in aggregate form after de-identification; the normative corpus, consisting of the trust provisions of Law No. 26702, SBS Resolution No. 1010-99 as amended, and applicable SBS circulars, retrieved from the SBS normative portal and Peru's official legal information system (SPIJ) with recorded consultation dates; prototype logs; retrieval traces; and expert annotations.

Gold standard. The gold standard is built by at least two, preferably three, lawyers with a minimum of five years of professional experience in trust or financial regulatory practice, who are external to the artefact development team and work from a written annotation guideline. Agreement is reported with Krippendorff's α (Krippendorff, 2018), with a minimum of $\alpha \geq 0.67$ and $\alpha \geq 0.80$ preferred, stated in advance. Disagreements are resolved through documented adjudication by a senior third expert. Annotator profiles will be reported in anonymised form. The gold standard represents defensible consensus positions rather than a single incontrovertible truth.

Evaluation design. Four conditions are compared on the same items: B0, a no-retrieval (closed-book) baseline; B1, standard RAG, the primary comparator; B2, the proposed artefact; and B3, current institutional practice (clauses drafted by legal staff under the usual procedure). B0–B2 use the same base model so that differences isolate the contribution of retrieval and of the verifiability layer. The unit of analysis is the drafting query together with its set of candidate clauses; outcomes are measured for each candidate and for the set as a whole (e.g., whether at least one suggestion is legally acceptable). Because generative outputs are stochastic, each query is run k times per system with fixed seeds where feasible, and variance across runs is reported. The number of queries and the value of k will be fixed by an a priori power analysis for paired proportions, assuming a two-sided significance level of 0.05 adjusted for multiple comparisons, a statistical power of 0.80, and a baseline hallucination rate for standard RAG informed by the 17–33% range reported for commercial legal RAG tools (Magesh et al., 2025); the calculation and its parameters will be recorded in the preregistration before data collection.

Table 4
Outcome and Process Measures for Artefact Evaluation

Measure	Role	Operationalisation
Attribution quality	Primary	Citation precision and recall (Gao et al., 2023) plus span-level attributability judged by annotators under the AIS criterion (Rashkin et al., 2023).
Hallucination rate	Primary	Proportion of generated propositions that are unsupported or misattributed (Ji et al., 2023), disaggregated by type.
Legal correctness / recommendation quality	Primary (non-inferiority)	Expert severity rubric applied per clause, aggregated by maximum severity so that a single decisive defect is not averaged away (Narayanan & Singh, 2026); includes contradiction with the governing framework.
Citation faithfulness	Secondary	Probe of post-rationalisation on a subsample (Wallat et al., 2025).

Measure	Role	Operationalisation
Outdated-source rate	Secondary	Proportion of citations to provisions not in force at the reference date.
Abstention rate and review effort	Secondary	Share of items where the system declines to recommend; review time and edits by legal officers relative to B3.
Accepted, edited, or rejected recommendations	Process	Indicator of human authority, relevance, and refinement needs.

Note. Author's own elaboration. Attribution (traceability) and legal correctness are measured as distinct constructs, as required by the tension identified in Deliverable 1.

Analysis. Systems are compared on the same items using mixed-effects logistic regression with random effects for item and run, complemented by paired bootstrap confidence intervals. Family-wise error across metric × baseline contrasts is controlled with the Holm–Bonferroni procedure. H3 is tested with a one-sided confidence interval against the margin Δ. Effect sizes and confidence intervals are reported alongside p-values.

Rigour and reproducibility. Before the summative evaluation, the hypotheses, outcomes, margin Δ, sample size, and analysis plan will be preregistered. Artefact versions, corpus snapshots, prompts, seeds, and annotation guidelines will be versioned so that each DSR iteration remains traceable.

Transferability. Transferability is claimed at the level of design principles, not of the instance. It rests on three mechanisms: an explicit abstraction step formulated with the anatomy of design principles (Gregor et al., 2020); validation by an expert panel of five to eight legal and compliance professionals from at least three SBS-supervised entities other than the host institution, drawn from both specialised fiduciary service companies and banks' trust units, who will assess the applicability of the principles to their own settings without access to institutional data; and an articulated account of the boundary conditions under which the principles hold. Generalisation is analytical, not statistical.

Feasibility boundary. The study evaluates one prototype, one institution, one clause family, and two to three primary outcomes. Synthetic stress tests may be used before human evaluation to probe misattribution and outdated-source failure modes, but they do not substitute for evidence on real, de-identified drafting tasks.

07. Ethical Considerations

Ethics approval will be obtained before recruiting annotators or processing any institutional data, through submission of the protocol to the competent research ethics

committee of the Universidad Nacional Mayor de San Marcos under the university's research regulations and the procedures of the doctoral programme. A written institutional authorisation from the host institution will define the scope of data use. Annotators and panel members will provide informed consent and will be bound by confidentiality agreements.

Legal regime. Clause text may contain information protected by banking secrecy under Law No. 26702 (art. 140) and personal data protected by Law No. 29733 and its current Regulation, approved by Supreme Decree No. 016-2024-JUS, which repealed the earlier Regulation (Supreme Decree No. 003-2013-JUS) cited in Deliverable 1. The use of AI will also observe Law No. 31814 and its Regulation (Supreme Decree No. 115-2025-PCM). Under that Regulation, AI systems not covered by the prohibited-use or high-risk categories are classified as acceptable-risk (art. 22.2). As an advisory clause-recommendation tool that neither manages critical infrastructure nor performs credit evaluation, the artefact is provisionally classified as acceptable-risk. Given its use in a supervised financial entity, however, the study voluntarily adopts the safeguards the Regulation sets for high-risk systems: human oversight by trained staff with the ability to stop, correct, or override the system (art. 31.4); an up-to-date record of the system's operating principles, data sources, and expected impacts (art. 31.1); and a documented impact assessment (art. 32). If the classification is uncertain, a formal consultation with the Secretariat of Government and Digital Transformation will be requested (art. 24.2).

Data protection by design. All contract text will be de-identified or pseudonymised before use; data minimisation will apply; identifiers will be kept separate from research records under restricted access; retention and deletion rules will be defined in advance. Processing will take place on-premise with a self-hosted model, so that no client data leaves the institution's controlled environment. If any external processing were unavoidable, a data-processing agreement and a data-protection impact assessment would precede it. No institutional data will be placed in public repositories; only aggregated results and de-identified artefacts approved by the institution will be published.

Human authority and stopping rules. The artefact is advisory: a qualified legal officer retains authority over any clause adopted. Any detected leakage of identifiable data, or any recommendation classified at the highest severity level during naturalistic evaluation, will trigger a predefined review and, where necessary, suspension of the evaluation.

Research integrity. The researcher's insider role and any conflict of interest will be declared. The use of generative AI tools in preparing thesis documents will be disclosed in accordance with the programme's integrity requirements, and all content will be verified by the researcher.

08. Expected Results

The expected results are stated as plausible outcomes rather than guaranteed findings. First, the project is expected to produce a functional, documented, verifiable clause-recommendation artefact whose architecture, verification rules, and human-review workflow are explicitly specified.

Second, the evaluation is expected to generate controlled evidence on whether an explicit verifiability layer is associated with changes in attribution quality, hallucination rate, and legal correctness relative to no-retrieval, standard RAG, and current-practice baselines.

Third, the study is expected to identify design trade-offs between what is citable and what is legally correct, between abstention and coverage, and between automation and human authority. These findings will support nascent, transferable design principles for verifiable generation in regulated domains. Negative, null, or heterogeneous findings will also be considered scientifically valuable because they reveal limits and failure modes; the thesis does not assume that the artefact must outperform the comparators.

Target venues include information-systems outlets receptive to DSR (DESRIST, *European Journal of Information Systems*, *Journal of the Association for Information Systems*), AI-and-law outlets (*Artificial Intelligence and Law*, ICAIL, JURIX), and NLP venues with tracks on attribution and factuality.

09. Timeline & Budget

Timeline. The work is planned across a thirty-six-month doctoral period, with explicit time for institutional authorisation, ethics approval, annotator calibration, iteration, and contingency rather than assuming that the first artefact version will succeed.

Table 5
Proposed 36-Month Research Timeline and Milestones

Period	Main activities	Milestone
Months 1–6	Consolidated protocol; PRISMA systematic review; institutional authorisation; ethics submission; requirements elicitation with the legal area; selection of the trust type and clause family; provisional AI-risk classification under Supreme Decree No. 115-2025-PCM.	Approved protocol, ethics submission, authorisation letter, registered review protocol.
Months 7–12	De-identification pipeline; versioned knowledge base of precedents and norms with periodic synchronisation; annotation guideline; annotator recruitment and calibration; architecture; prototype alpha (query autocompletion, retrieval, and generation).	Prototype alpha, guideline, calibration agreement.
Months 13–18	Verifiability layer (proposition-level attribution, validity check, abstention); review interface; formative evaluation; stress tests; expert consensus on the margin Δ; power analysis; voluntary AI impact assessment; preregistration on OSF.	Evaluation-ready artefact, impact assessment, and preregistration.
Months 19–24	Dated snapshot of the knowledge base; gold-standard construction; summative evaluation of B0–B2 with repeated runs; collection of B3 data.	Gold standard with reported α; evaluation dataset.
Months 25–30	Statistical analysis; naturalistic demonstration with legal officers; expert panel from other supervised entities; refinement iteration.	Evaluation results, panel report, refined artefact
Months 31–34	Abstraction of design principles; manuscripts for IS and AI-and-law venues; integration of ethics and limitations findings.	Draft publications and contribution chapter.
Months 35–36	Thesis integration, final quality review, documentation, and defence preparation.	Final dissertation and defence package.

Note. Author's own elaboration. Milestones will be aligned with the formal stages established by the UNMSM doctoral regulations (thesis-plan approval, periodic progress reviews, and defence), whose calendar dates will be set with the programme coordination.

Preliminary budget. Amounts are not estimated at this stage: each item will be costed from the indicated source once access, institutional in-kind support, and ethics conditions are confirmed. The budget excludes tuition and the researcher's own labour. Foreign-currency quotations will be converted at the BCRP exchange rate on the quotation date.

Table 6
Preliminary Budget for the Design, Development, and Evaluation of the Artefact

Item	Amount (S/)	Rationale and source
Legal annotators (2–3 lawyers)	30 000	Gold-standard construction and calibration. Source: quotations from at least two specialists (fee × estimated hours); reference: ponteencarrera.pe.

Item	Amount (S/)	Rationale and source
Senior adjudicator and expert panel	7500	Adjudication of disagreements and validation of design principles. Source: direct quotations.
Computing infrastructure (on-premise)	18000	Self-hosted model inference. Source: host-institution infrastructure (in-kind, valued if applicable) or local vendor quotations.
Secure data management	1500	Encrypted storage, backups, access control. Source: vendor quotations or institutional in-kind support.
Literature access	1000	Articles not covered by UNMSM subscriptions. Source: UNMSM library services.
Dissemination	6000	Conference registration and publication fees. Source: official fee pages of target venues.
Contingency (15% of subtotal)	9600	Re-annotation, re-testing, or replacement costs.
Preliminary total	73600	Subtotal plus contingency.

Note. Author's own elaboration. The budget identifies cost items and their sources; amounts will be entered once quotations and in-kind support are confirmed.

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